

Sheth L.U.J & Sir M.V College Of Science
Subject :- Data Analysis With SAS/SPSS/R
Practical No 13

Aim :- Performing linear regression analysis using lm() (R).

```
Console Terminal Background Jobs
R - R452 - ~/ -
> # Load dataset
> data <- read.csv("data.csv")
>
> # Take top 5 rows only
> data_top5 <- head(data, 5)
>
> # View data
> print(data_top5)
  X Year GDP_Current_USD GDP_Growth_Percent GDP_Per_Capita_USD Inflation_Rate_Percent Official_Exchange_Rate Population_Total
1 0 1960 4199134390 NA 195.5775 9.8224108 75.75 21470434
2 1 1961 4426949095 10.390918 200.0815 3.1625644 75.75 22125726
3 2 1962 4693566416 7.909268 205.7868 0.7213206 75.75 22807907
4 3 1963 4928628018 7.042075 209.5594 0.3718496 75.75 23519005
5 4 1964 5379845648 8.475228 221.7317 3.8144896 75.75 24262859
  Population_Growth_Percent Exports_USD Imports_USD Trade_Percent_GDP Industry_Value_Added_Percent_GDP
1 NA 581541416 730164939 31.23754 25.74758
2 3.006418 607915177 702251188 29.59524 27.82821
3 3.036627 680443018 642016254 28.17600 29.54496
4 3.070154 755608235 614102504 27.79091 30.86295
5 3.113793 882202285 878548558 32.72865 31.40206
>
> # Select numeric columns
> numeric_data <- data_top5[apply(data_top5, is.numeric)]
>
> # Assign variables
> Y <- numeric_data[[1]]
> X <- numeric_data[[2]]
>
> # Linear regression model
> model <- lm(Y ~ X)
>
> # Model summary
> summary(model)

Call:
lm(formula = Y ~ X)

Residuals:
    1     2     3     4     5 
-3.315e-14  3.315e-14  1.661e-14 -7.428e-17 -1.654e-14 

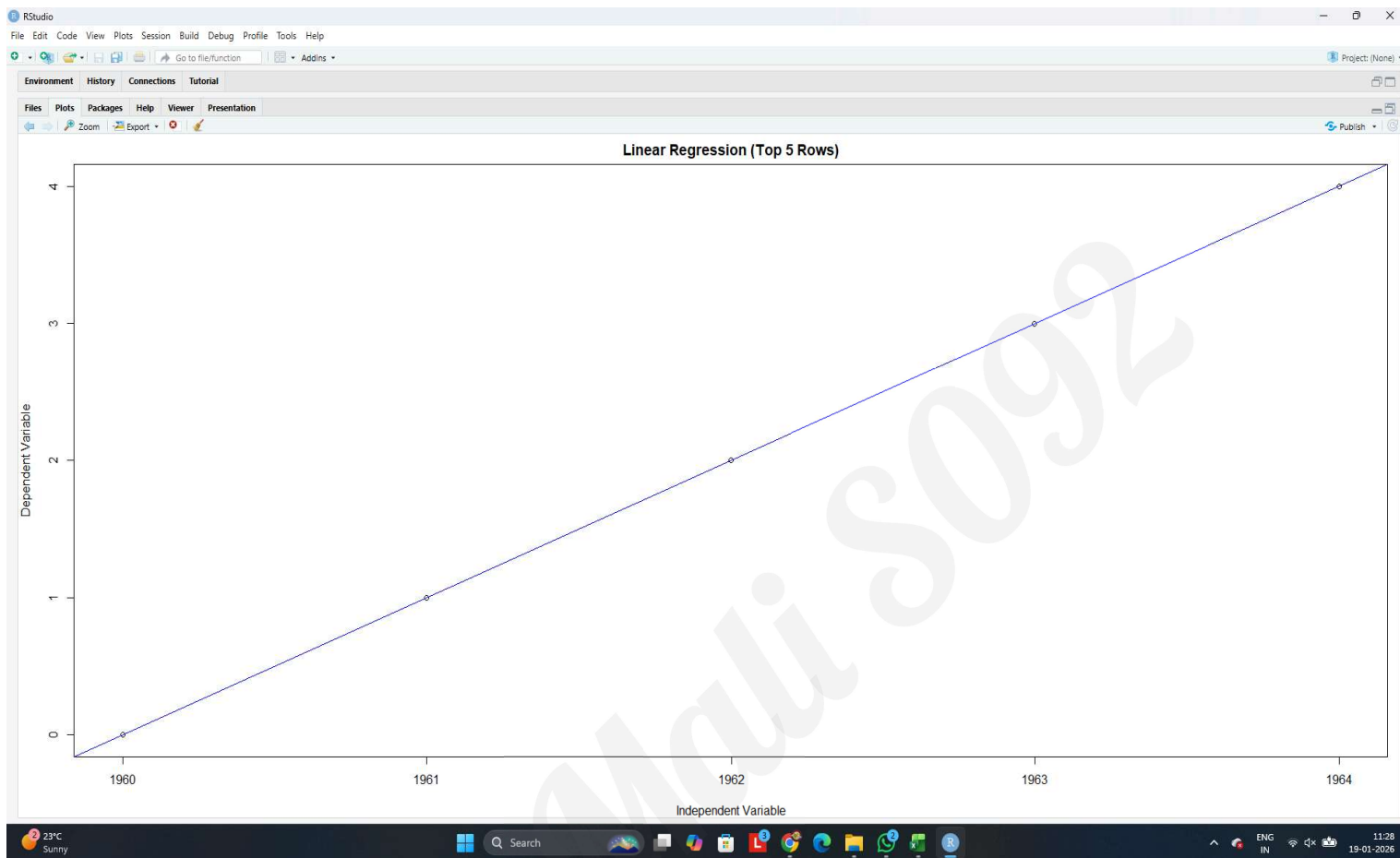
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.960e+03  1.877e-11 -1.044e+14 <2e-16 ***
X              1.000e+00  9.568e-15  1.045e+14 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.026e-14 on 3 degrees of freedom
Multiple R-squared:  1, Adjusted R-squared:  1
F-statistic: 1.092e+28 on 1 and 3 DF, p-value: < 2.2e-16

> # Fix plot margin error
```

Name :- Vinod Mali
Roll No :- S092

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